AUTOMATICALLY COMBINING CONCEPTUAL MODELS USING SEMANTIC AND STRUCTURAL INFORMATION

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ABSTRACT

A conceptual model is a necessary precursor to a simulation model. It defines the problem space by listing relevant concepts and it identifies salient mechanisms by specifying which concepts are related. In this paper, we focus on conceptual models expressed as causal or 'cognitive' maps, in which concepts form nodes and their relationships are directed, weighted edges specifying causal strengths. When modeling complex social systems, participants may provide their perspectives through individual causal maps. To create a simulation model, these maps need to be aggregated into a coherent conceptual model. Two challenges arise: nodes may have different names although participants ascribed the same meanings (i.e., linguistic variability) and disagreements on causal strengths need to be reconciled. Although other fields have long proposed algorithms to aggregate knowledge bases (e.g., ontology matching), there is a paucity of solutions for causal maps. In this paper, we propose a solution for causal maps that leverages recent advances in graph matching. We demonstrate the feasibility and potential of our approach on a case study with $n = 22$ maps.

Keywords: Causal Maps, Conceptual Model, Mental Model, Utility Matrix, WordNet.

1 INTRODUCTION

In Participatory Modeling (PM), the conceptual design of a model is shaped by the perspectives of participants who may be experts, stakeholders, or community members. By combining these qualitative perspectives, a model can be created even when there is a limited quantitative evidence available and/or high uncertainty about the evidence base. There are various processes to elicit knowledge from individuals, which affect the choice of processes to combine knowledge across individuals. In this paper, we are interested in knowledge that is elicited in the form of concept maps, also known as cognitive maps (Voinov et al. 2018). Elicitation methods resulting in this formal representation include a semi-structured interview of individual participants either with a trained facilitator (Furman et al. 2021, Firmansyah et al. 2019, Papageorgiou and Kontogianni 2012) or a software (Reddy et al. 2019), text mining from transcripts (Pratt et al. 2013), or group-based modeling (Tegarden and Sheetz 2003).

Two challenges arise when attempting to combine the many conceptual models elicited from individuals into a single model that can be operationalized for simulations. The main challenge is *linguistic variability*: participants may use different words which, in a given context, have the same meaning. The linguistic

ANNSIM'21, July 19-22, 2021, Fairfax, VA, USA; ©2021 Society for Modeling & Simulation International (SCS)

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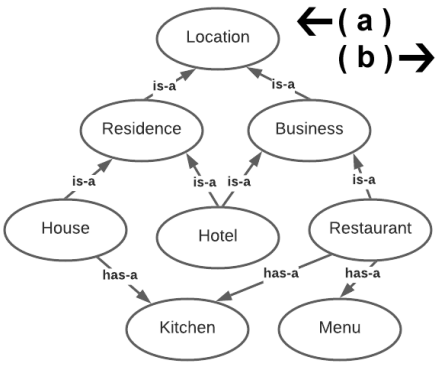

Method	Metrics Used		
	Semantic	**Syntactic**	**Structural**
OMM	WordNet Synonym Set for each part-of-speech	N/A	N/A
DSSim	WordNet Synonym Set	Levenshtein Distance	Leaf Node Set
FuzzyAlign	WordNet Lexical Position	Jaccard Coefficient	Hierarchical and Property types
MapSSS	Google Research Synonyms and Translations API	Levenshtein Distance	Existing Neighbor Match

Figure 1: Sample ontology of locations (a) and similarity metrics used to aggregate ontologies (b).

variability within conceptual mapping has several consequences: time and effort will be required to align linguistics, potential erroneous outcomes can compromise the validity of the model, and ambiguous meanings could lead to incorrect recommendations based on misinterpretations. This challenge is documented in research[1] on the interoperability of conceptual models (Tolk and Muguira 2003). Linguistic variability can be avoided when participants are only allowed to choose terms from a pre-established set (Gray et al. 2015) or when they are only asked to share their perspective on the causal strength on a pre-established map (Giabbanelli 2014). In such cases, it becomes straightforward to aggregate maps, which allows to conduct large-scale experiments in collective intelligence by allowing many participants to shape a model (Aminpour et al. 2020, Gray et al. 2020). However, to preserve the richness of qualitative information elicited from participatory modeling, participants can usually choose their terms. This challenge is illustrated through algorithms and software that assist analysts in manually aligning conceptual models in the face of linguistic variability (Giabbanelli et al. 2019, Gupta et al. 2018). A secondary challenge is to solve *disagreements* or 'inconsistencies' regarding the mechanisms of a system. For example, two participants may mention that wearing face masks decreases the risk of transmitting COVID-19, but they may disagree on the intensity of this relationship (i.e., 'causal strength'). This may be solved through by averaging or augmenting a cognitive map with information granules (i.e., 'granular computing') such as fuzzy logic, rough set theory, or combinations thereof (Nápoles et al. 2016, Falcon et al. 2019).

Aggregation algorithms have been developed in other fields modeling knowledge through network structures, such as ontologies (Figure 1a) (Otero-Cerdeira et al. 2015). By automating the aggregation process, researchers can eliminate inconsistencies and solve discrepancies in terminology, thus allowing aggregation decisions to be made logically based on properties of the concepts and relations themselves. Such computer-driven aggregation approaches are repeatable at scale, thus allowing for a broad participation of individuals and/or very large maps. However, there is a paucity of methods to combine causal maps as techniques from other fields are generally not directly applicable to causal maps. For instance, ontology aggregation methods take into account concept classes and hierarchical relationships, neither of which are present in causal maps.

At a high level, a common theme across aggregation methods is to use a combination of *Semantic Similarity*, in which linguistics are used to compare meaning, and *Structural Similarity*, in which the positions of concept nodes' within a network are compared. In this paper, we propose to complement these two techniques with a *utility matrix* to match terms from different participants. This is a recent technique from the field of Graph Matching that has never been applied to semantic similarity. Our main contributions are twofold:

[1]Note that related research areas such as *semantic composability* (Szabo and Teo 2009) operate at the level of *simulation* runs rather than conceptual models. Formal approaches for conceptual models (Gore et al. 2014) are also inapplicable here, since they rely on mathematical properties (e.g., constraints) that are unavailable for concept maps.

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- We demonstrate the effectiveness of complementing semantic and structural similarities with a utility matrix to combine conceptual models formalized as causal maps.
- We apply our proposed aggregation procedure to a case study pertaining to a model of suicide, which has been under-researched in the simulation literature.

The remainder of this paper is structured as follows. In section 2, we briefly summarize current aggregation techniques for contextualized networks, drawing from fields such as fuzzy cognitive mapping and ontologies. In section 3, we describe our proposed algorithm. Then, we apply it to our case study in section 4. Finally, we explain how results from the case study demonstrate the value of a utility matrix.

2 BACKGROUND

2.1 Aggregating Maps

The ability to combine models representing the same system is a common requirement in conceptual modeling. This has led to the development of aggregation techniques for network-based models as diverse as ontologies, fuzzy cognitive maps, or process models (La Rosa et al. 2013). A process to analyze multiple Ontologies within the same context is *Ontology Matching*, which finds correspondences between semantically related entities of ontologies (Figure 1b). These alignments enable the knowledge and data expressed with respect to the matched ontologies to interoperate (Otero-Cerdeira et al. 2015). One technique for finding these correspondences is the *Ontology Matching Method* (OMM), which relies exclusively on semantic comparisons (Lee and Lee 2015). The OMM assigns a weight to each part-of-speech (noun, verb, adjective, etc) and compares two concepts by computing a WordNet similarity score for each of their part-of-speech word subsets. Another method is *DSSim*, in which each concept ("query graph fragment") from Ontology A is compared against Ontology B, where all concepts in B are extended to include their synonyms (Nagy et al. 2006). Then, the query fragment is aligned to a concept in B based on a combination of Levenshtein distance[2], set of synonyms, and set of leaf nodes. Next there is *FuzzyAlign*, which considers the number of shared Wikipedia pages returned from searching concepts u and v, the number of shared lexicographic roots of u, v in WordNet, and whether u, v are of the same type in the Ontology (Fernández et al.). Lastly, there is *MapSSS*, which considers the Levenshtein distance between u, v, the neighbors of u, v that have already been matched, and the semantic similarity score of u, v when using the Google Research API (Cheatham 2011). We note that these approaches are not directly applicable for causal maps. For example, the FuzzyAlign method relies on comparing the 'type' of two concepts, which is only applicable to ontological networks, and the MapSSS method assumes that concepts whose neighbors have been matched together both have the same hierarchical relationships with those neighbors.

2.2 Graph Matching

A similar process to Ontology Matching is that of *Graph Matching*, which seeks to find a pairwise correspondence between the vertices of two graphs. The problem of *graph isomorphism* is to find a perfect matching between two graphs, but such a matching is rarely possible. We are thus interested in an *inexact graph matching*. The *graph edit distance* (GED) is a method that quantifies the similarity between two graphs by identifying a sequence of transformations with the lowest total cost in terms of vertex/edge additions, substitutions, and deletions. Intuitively, the GED shows us which operations are sufficient to transform one graph into another, and it summarizes differences between these graphs through these operations. Finding

[2]The Levenshtein distance is a syntactic distance metric that evaluates the number of edit operations (substitutions, insertions, and deletions of symbols) needed to transform one string into another. (Konstantinidis 2005)

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an optimal matching is a quadratic assignment problem, with a cost of $O(m^n)$ for graphs of m and n vertices. Given this extreme computational cost, optimal solutions can only be computed in very small graphs.

For large practical applications, one approach is to approximate the by generating a *cost matrix* C with rows representing vertices from one graph and columns representing vertices from the other graph. Each position C_{ij} is evaluated based on a cost function of matching vertices i and j (Gao et al. 2010). The lowest cost value in the matrix is selected, those vertices are removed, and the process continues until all vertices have been matched. Alternatively, a recent method computes a *row utility matrix* C_{row} by visiting each vertex in the first graph and computing the relative cost of matching that vertex with every vertex in the second graph, where each relative cost is computed as follows:

$$C_{row} = \frac{row_{max} + row_{min} - 2C_{ij}}{row_{max} - row_{min}} \tag{1}$$

where C_{ij} is a cost function between vertices i and j (Riesen et al. 2018). Then, the same procedure is performed in the opposite direction to obtain the *column utility matrix* C_{col}. Finally, the matrices are combined to obtain a *utility matrix* U. We can obtain a graph matching by selecting the lowest cost pair in the utility matrix, removing those vertices, and continuing until all vertices have been matched. This has been shown to be a strong approximation of graph matching since the relative costs with respect to both rows and columns are considered when matching vertices.

For example, consider the cost matrix C in Figure 2. A simple approximation would give us the matching $M = \{u_1 \to v_1, u_2 \to v_2, u_3 \to v_3\}$ with a cost of 14. Alternatively, by combining the row and column matrices to get a utility matrix U, we obtain the matching $M = \{u_1 \to v_1, u_2 \to v_3, u_3 \to v_2\}$ with a cost of 9.

$$C = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} 1 & 2 & 3 \\ 4 & 3 & 3 \\ 3 & 5 & 10 \end{pmatrix} \end{array} \quad U_{row} = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 1 \\ 1 & \frac{3}{7} & -1 \end{pmatrix} \end{array} \quad U_{col} = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 \\ -1 & \frac{1}{3} & 1 \\ -\frac{1}{3} & -1 & -1 \end{pmatrix} \end{array} \quad U = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} 2 & 1 & 0 \\ -2 & \frac{4}{3} & 2 \\ \frac{2}{3} & -\frac{4}{7} & -2 \end{pmatrix} \end{array}$$

Figure 2: Evaluating cost of graph matching with a Utility Matrix.

We note that this utility matrix developed for graph matching is re-purposed in our research to match words between the bodies of text inferred from conceptual models. The application discussed in section 3.1.1 thus builds on the notation introduced here.

2.3 Foundations of our Case Study: Causal Mapping in Suicide Research

Our case study pertains to suicide in children and adolescents, thus we briefly summarize research on conceptual modeling in this area. Ten years ago, a map-based model categorized whether a text was a suicide note. It focused on elements of the text rather than high-level social concepts, hence its map is formed of common words such as ‘me’, ‘wanted’, ‘tried’, or ‘they’ (White and Mazlack 2011). In parallel, the cultural theory and model of suicide came into prominence, which later inspired detailed maps containing factors such as ‘perceived self-worth’ (Chung 2016). Other models have captured the steps of suicide, from ideation and planning to attempt and fatality (Page et al. 2017). Most recently, an ontology was presented which articulates how 297 factors (e.g., emotional neglect, witnessing neighborhood violence, or parental separation) are connected and ultimately lead to adverse childhood experiences, which constitute a risk factor for suicide (Brenas et al. 2019). In an ongoing project, we have aimed at combining these strands of literature

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(cultural theory, steps of suicide, adverse childhood experiences) by transforming interviews of $n = 22$ subject matter experts into maps related to various facets of suicide. *These previously collected maps serve are used as a case study in this paper to demonstrate our algorithm.*

3 METHODS

3.1 Overview of our Experimental Design and Method

Key steps of our approach are shown in Figure 3. The first step towards our aggregation algorithm is to measure *semantic similarity*. In our causal maps, nodes have textual labels which are essential to detect meaningful overlap in their definitions. We will use a common factor similarity calculation from Ontology matching as a baseline approach and compare it with our proposed method that uses a utility matrix. Next, we will compute the *structural similarity* scores between nodes based on shared characteristics revealed through the nodes' locations in the causal maps. This score is based on the proportion of shared neighbors as source nodes, target nodes, and from edges of positive and negative signs. Finally, we will compute a *Linking Metric* as a sum of the semantic and structural similarity scores and store these metric values in a similarity matrix. This matrix will be used to aggregate nodes across maps sequentially, starting with the pairs of highest similarity, and continuing until a threshold similarity has been crossed. To merge a pair of nodes, we select the highest similarity score from the matrix, and pick either node in the pair to inherit all the neighbors of the other node which is then deleted. After merging two nodes, some values in our similarity matrix will need to be updated. To avoid the computational cost of recomputing every value, we update values along the rows and columns of the two nodes involved in the most recent merge. The scripts implemented our approach are openly available on a third-party repository at https://osf.io/hpz7c/.

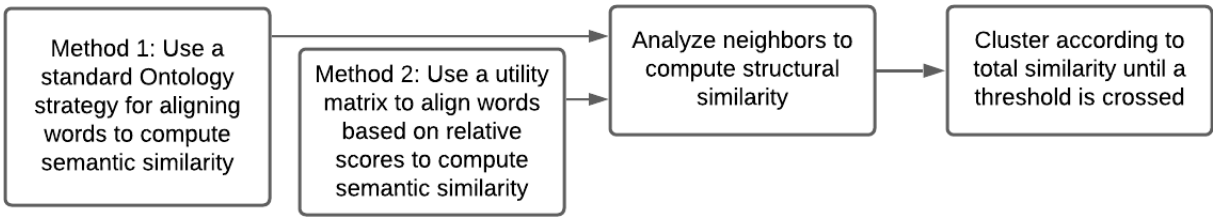

Figure 3: Steps for aggregating causal maps.

3.1.1 Semantic Similarity

Before interpreting the semantic meaning of textual labels, we preprocess them to filter out meaningless content and standardize formatting. We remove capitalization of letters, punctuation occurring between words, and stop words such as "the" or "and", which do not provide distinction between labels. Then, each label is *tokenized*, that is, converted into a list of word tokens. If node labels contain information that cannot be interpreted automatically (e.g., acronyms, numbers), we use custom transformations (described below).

Early Ontology Matching techniques used the Jaccard Similarity to evaluate semantic similarity, while recent methods use background knowledge sources including internet or database searches based on a description attached to each node. After obtaining a long format description, these methods still compute a Jaccard Similarity, as it is an established technique for computing semantic similarity (Cross 2018). To compute the Jaccard Similarity, many papers use the WordNet lexicographical database to evaluate the similarity of word pairs. Each word in the first factor is matched with its most similar word in the second factor. Once no more matches can be made, the average metric across all word matches is given as the total semantic similarity. Because our causal maps do not contain descriptions for factors, our baseline semantic similarity

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method obtains word pair similarity from WordNet. To compute a semantic similarity score between two factors, we align their labels on words that have the most similar sets of synonyms. To evaluate the distance between any two words, we use the WuPalmer and Shortest Path algorithms in WordNet as a cost function. The WuPalmer algorithm compares the depth of words and their synonym sets in the WordNet taxonomy, and the Shortest Path algorithm measures the shortest distance between words in a network of synonyms.

A second approach we use when aligning words is to implement a utility matrix ordering, in which matching priority is evaluated with respect to both rows and columns before. We use a WordNet similarity metric to construct a cost matrix comparing synonym sets from each factor, and evaluate relative costs along every row and column to obtain a utility matrix. Finally, we select an alignment of words according to the utility matrix and use the sum of scores as the total semantic similarity between the two factors.

While WordNet excels at comparing nouns, other word types occur in our labels when computing the semantic similarity. We address this limitation using a custom procedure to transform the part-of-speech of all adjectives, verbs, adverbs, and satellite adjectives, or adjectives that have meaning in a particular context such as "socio" or "undiagnosed". We first attempt to convert each word to a noun by looking for noun synonyms of the word, given by WordNet. In cases where no such synonym is available, we search the synonyms of each synonym, and so on recursively. When multiple noun synonyms are present, we select the one with smallest Levenshtein distance from the original word.

Another challenge occurs when a noun is not indexed in the WordNet similarity algorithms. In such cases, we search for alternative nouns through a similar part-of-speech transformation until we find a satisfying noun. This process can become computationally intensive, so we save similarity scores for a pair of words once they have been computed and re-use them any time we encounter the same pair of words.

3.1.2 Structural Similarity

The second measure of similarity between nodes relies on their corresponding edges in the causal map. More specifically, nodes with similar sets of neighbors are considered to be more similar themselves. Because edges in our causal maps are both signed and directed, we must compare the four sets of source/target and positive/negative neighbors separately. To compute the structural similarity between nodes u and v, we use the Jaccard Coefficient as follows:

$$S_{stuctural} = \frac{|s^+(u) \cap s^+(v)| + |s^-(u) \cap s^-(v)| + |t^+(u) \cap t^+(v)| + |t^-(u) \cap t^-(v)|}{|s^+(u) \cup s^+(v)| + |s^-(u) \cup s^-(v)| + |t^+(u) \cup t^+(v)| + |t^-(u) \cup t^-(v)|} \tag{2}$$

where $s^+(u), s^-(u)$ are the sets of positive and negative source neighbors of u, and $t^+(u), t^-(u)$ are the sets of positive and negative target neighbors of u.

3.2 Parameters

For a fair experimental comparison, optimal values need to be established for the parameters of our aggregation algorithm listed in Table 1. The *preprocess* specifies whether the algorithm should use our custom translations (e.g., transform the acronyms which we listed) or only perform standard pre-processing (Translated or Raw). The *metric* specifies the WordNet algorithm that will be used to measure semantic similarity (WuPalmer or Shortest Path). The *alignment* specifies whether to use a utility matrix for aligning words or a standard similarity matrix when computing semantic similarity – this parameter allows us to switch between our proposed approach and a baseline from current practices (Utility or Standard). The α dictates the relative weight of the semantic and structural similarity scores as follows:

$$S(i,j) = \alpha S_{semantic}(i,j) + (1-\alpha) S_{structural}(i,j) \tag{3}$$

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Table 1: Parameters used in the aggregation of causal maps.

Parameter	**Notation**	**Description**	**Possible Values**
Pre-processing	preprocess	Whether to perform additional translations or only raw preprocessing	$\{translated, raw\}$
Metric	metric	The WordNet algorithm to use for comparing individual words	$\{Wu-Palmer, Shortest\}$
Alignment	alignment	Whether to use a utility or standard similarity matrix when computing semantic similarity	$\{utility, standard\}$
Alpha	α	The proportion of total similarity that will come from semantic similarity	$[0.0, 1.0]$
Threshold	threshold	The minimum similarity score that must be present in order to form a cluster	$[0.0, 1.0]$

where $S(i, j)$ is the total similarity between nodes i and j. Finally, the *threshold* parameter is used to stop aggregating after a desired level of similarity has been reached. That is, once all remaining pairs of nodes have a similarity less than *threshold*, the algorithm will finish execution.

3.3 Evaluation

We perform the algorithm on the $n = 22$ causal maps described in section 2.3. Our aggregation approach is akin to an *agglomerative clustering* as we assign nodes across maps to an initial cluster and sequentially merge them into one map as long as we are over the desired similarity threshold. To track how the final map was obtained, we record the final cluster assigned to each node. This cluster assignment will be essential for the evaluation, as it will be compared to a ground truth dataset for accuracy (see below). Our experiment is repeated for every combination of our parameters: metric, preprocess, α, utility, and threshold.

In our previous project, the $n = 22$ causal maps were manually combined into one. This process was performed by two modelers with guidance from two subject matter experts; several meetings took place to repeatedly identify and resolve potential duplicates. Modelers created a thesaurus to document how factors were handled. Node labels listed individually in the thesaurus indicate that they were not similar to other concepts and hence remained separate in the aggregate map. For example, an interviewee may have been the only one to mention 'substance abuse' hence this factor appears as such in the final map. Conversely, node labels with manually identified synonyms indicate that concepts across interviewees were deemed equivalent and merged. For instance, one interviewee's map may contain 'access to lethal means' while another may have 'access to means of suicide': these two equivalent concepts would be combined into a single concept in the aggregate map, with a thesaurus entry indicating the name retained for the combination.

We use this thesaurus to verify whether a node was listed as having any similar labels: if it was, then 'merge' was the correct action in the ground-truth dataset; if it was not, then 'do not merge' was the ground-truth action. These binary ground-truth actions are then compared against the actions taken by our algorithm. If the algorithm places a node in a cluster and 'merge' was the correct action, then we count it as a true positive. If the algorithm instead decided to keep a node by itself and 'do not merge' was the correct action, then we count it as a true negative. Consequently, our evaluation is akin to a binary classification and produces a *confusion matrix*, which records true positives (tp), false positives (fp), true negatives (tn), and false negatives (fn). The confusion matrix is useful in recognizing areas of strong performance in our algorithm, and can be used to make parameter adjustments. Most importantly, we use it to evaluate the effectiveness of our proposed utility matrix for computing semantic similarity. Since our data has a notable class imbalance,

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we use the F-1 score as performance indicator for our algorithm. This score is derived from the confusion matrix as F-1 $= 2 \times (p \times r)/(p+r)$, using the *precision* $p = tp/(tp+fp)$ and the *recall* $r = tp/(tp+fn)$.

4 RESULTS

We now provide the results of our case study evaluation as described in the previous section. First, we detail the particular semantic translations that we made made for experiments that used the *translation* parameter and show the set of factors that were still unrecognized after making these changes. Then, we explain how values were selected for α and *threshold* and we visualize the results across the search space of all 5 parameters using confusion matrices. Due to the page limits, we present one of these matrices in Figure 4; interactive visualizations of all matrices are provided online at https://public.tableau.com/profile/alex.freund#!/vizhome/CausalMapAggregation/HeatMaps?publish=yes. Finally, we analyze the results. Note that our experiment is a binary classification problem, so we must select an appropriate metric to interpreting our results. In our thesaurus, 134 factors were clustered (37.1%) and 227 were not clustered (62.9%).

4.1 Raw Pre-processing vs. Pre-processing Including Translations

Our $n = 22$ maps contained a total of 361 causal map factors, forming 626 unique words. After standard preprocessing (i.e., punctuation removal, consistent capitalization, removal of stopwords), we obtained 450 words. We set aside this cleaned dataset for experiments where $preprocess = raw$. A copy of this dataset was further transformed by applying manual translations (thus constituting the $preprocess = translate$ dataset) as follows. We first identified 2 factors that contained numbers and translated them entirely: Age 10 - 17 $\rightarrow$ Teenager, Younger age (<4) $\rightarrow$ Toddler. For the remaining 359 factors, which account for 446 unique words, we automatically tracked the words that could not recognized by the WordNet API, giving us a subset of 35 such words. Of these words, we manually translated 29 to handle spelling errors, acronyms, and jargon[3], while the remaining 6 were prepositions and prefixes that were still unrecognized by WordNet[4]. We considered this latter set of 6 words to be irrelevant in the context of our case study and removed them. In total, our algorithm flagged 39 of the 450 preprocessed words for us to translate manually.

4.2 Search Space

Besides the *preprocess* parameter, we also divided experiments according to the binary *metric* and *alignment* parameters, resulting in $2^3 = 8$ parameter combinations. Within each of these, we ran our algorithm separately across numerical values of α and *threshold*. To determine values for α and *threshold*, we first performed our entire experiment using $\alpha = \{0.0, 0.1, 0.2, ..., 1.0\}$ and *threshold* $= \{0.0, 0.1, 0.2, ..., 1.0\}$. This revealed positive skewness of our similarity scores at $s \leq 0.45$, leading to every factor getting clustered in some configurations when *threshold* was too small. We also observed very few scores above $s = 0.70$. Thus, we conducted our experiment with higher granularity using $\alpha = \{0.00, 0.05, 0.10, ..., 1.00\}$ and *threshold* $= \{0.450, 0.475, 0.500, ..., 0.700\}$, resulting in $8 \cdot 21 \cdot 11 = 1,848$ algorithm runs.

In total, we produced 8 confusion matrices, that is, one per combination of the binary parameters (*preprocess*, *metric*, *alignment*). In each confusion matrix, we use a heatmap to show how results depend on the $21 \cdot 11 = 231$ combinations of values for α and *threshold*. One such confusion matrix is shown in Figure 4.

[3]The 29 translations are provided online on https://osf.io/hpz7c/, as 'Manual Translations' under ANNSIM2021.

[4]Words still unrecognized by WordNet after making translations: via, non, within, re, among, co

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Action taken by the algorithm (i.e., 'prediction')

Action taken by modelers (i.e., experimental 'ground truth')

Merge — *True positives*

Alpha	0.45	0.475	0.5	0.525	0.55	0.575	0.6	0.625	0.65	0.675	0.7
0	11	11	9	7	7	7	7	7	6	6	4
0.05	12	10	8	8	8	8	8	6	6	6	5
0.1	12	10	8	8	8	8	8	6	6	6	5
0.15	12	10	8	8	8	8	8	6	6	6	5
0.2	13	10	8	8	8	8	8	6	6	6	5
0.25	13	13	8	8	8	6	6	6	5	5	5
0.3	15	13	8	8	6	6	6	5	5	5	5
0.35	19	15	9	7	6	6	5	5	5	5	5
0.4	23	17	11	6	6	6	6	5	5	4	4
0.45	28	21	15	8	6	6	5	5	4	4	4
0.5	38	27	22	12	8	6	5	4	4	4	4
0.55	56	45	31	17	12	10	5	4	4	4	3
0.6	79	58	50	30	17	12	8	4	4	3	3
0.65	93	78	70	50	31	21	12	7	3	3	3
0.7	106	92	81	77	59	42	29	12	6	3	0
0.75	115	106	95	82	76	70	53	35	16	4	0
0.8	132	117	105	96	90	80	73	58	37	21	7
0.85	133	126	118	108	98	92	83	70	69	44	24
0.9	134	134	126	121	108	101	95	85	73	69	58
0.95	133	133	133	127	122	108	98	95	89	75	68
1	133	133	133	133	129	122	114	103	92	87	78

Do not merge — *False negatives*

Alpha	0.45	0.475	0.5	0.525	0.55	0.575	0.6	0.625	0.65	0.675	0.7
0	123	123	125	127	127	127	127	127	128	128	130
0.05	122	124	126	126	126	126	126	128	128	128	129
0.1	122	124	126	126	126	126	126	128	128	128	129
0.15	122	124	126	126	126	126	126	128	128	128	129
0.2	121	124	126	126	126	126	126	128	128	128	129
0.25	121	121	126	126	126	128	128	128	129	129	129
0.3	119	121	126	126	128	128	128	129	129	129	129
0.35	115	119	125	127	128	128	129	129	129	129	129
0.4	111	117	123	128	128	128	128	129	129	130	130
0.45	106	113	119	126	128	128	129	129	130	130	130
0.5	96	107	112	122	126	128	129	130	130	130	130
0.55	78	89	103	117	122	124	129	130	130	130	131
0.6	55	76	84	104	117	122	126	130	130	131	131
0.65	41	56	64	84	103	113	122	127	131	131	131
0.7	28	42	53	57	75	92	105	122	128	131	134
0.75	19	28	39	52	58	64	81	99	118	130	134
0.8	2	17	29	38	44	54	61	76	97	113	127
0.85	1	8	16	26	36	42	51	64	65	90	110
0.9	0	0	8	13	26	33	39	49	61	65	76
0.95	1	1	1	7	12	26	36	39	45	59	66
1	1	1	1	1	5	12	20	31	42	47	56

Do not merge — *False positives*

Alpha	0.45	0.475	0.5	0.525	0.55	0.575	0.6	0.625	0.65	0.675	0.7
0	66	64	64	46	44	43	42	40	39	38	38
0.05	63	63	46	44	42	41	40	40	39	37	36
0.1	63	62	46	44	42	41	40	40	39	37	36
0.15	63	58	46	44	42	41	40	40	39	37	36
0.2	63	56	46	44	42	41	40	40	39	37	36
0.25	65	55	42	41	40	40	39	37	36	35	33
0.3	67	55	44	40	40	39	37	36	35	33	32
0.35	69	57	45	40	39	38	37	35	33	32	30
0.4	71	61	47	38	36	34	33	32	30	30	28
0.45	78	63	53	38	34	33	32	30	30	28	26
0.5	84	67	58	44	35	33	32	30	28	26	20
0.55	99	83	63	50	44	37	32	30	22	20	17
0.6	128	99	80	61	52	39	35	26	24	17	16
0.65	157	128	110	85	68	53	34	29	25	20	12
0.7	179	164	134	106	85	64	51	34	25	20	14
0.75	204	180	163	135	105	92	67	58	33	20	14
0.8	224	211	182	163	145	109	90	65	51	37	21
0.85	226	224	208	184	163	151	121	92	84	48	38
0.9	226	225	220	205	183	164	153	132	100	84	60
0.95	227	227	227	221	206	187	160	150	132	108	91
1	227	227	227	227	222	200	186	169	153	139	121

Do not merge — *True negatives*

Alpha	0.45	0.475	0.5	0.525	0.55	0.575	0.6	0.625	0.65	0.675	0.7
0	161	163	163	181	183	184	185	187	188	189	189
0.05	164	164	181	183	185	186	187	187	188	190	191
0.1	164	165	181	183	185	186	187	187	188	190	191
0.15	164	169	181	183	185	186	187	187	188	190	191
0.2	164	171	181	183	185	186	187	187	188	190	191
0.25	162	172	185	186	187	187	188	190	191	192	194
0.3	160	172	183	187	187	188	190	191	192	194	195
0.35	158	170	182	187	188	189	190	192	194	195	197
0.4	156	166	180	189	191	193	194	195	197	197	199
0.45	149	164	174	189	193	194	195	197	197	199	201
0.5	143	160	169	183	192	194	195	197	199	201	207
0.55	128	144	164	177	183	190	195	197	205	207	210
0.6	99	128	147	166	175	188	192	201	203	210	211
0.65	70	99	117	142	159	174	193	198	202	207	215
0.7	48	63	93	121	142	163	176	193	202	207	213
0.75	23	47	64	92	122	135	160	169	194	207	213
0.8	3	16	45	64	82	118	137	162	176	190	206
0.85	1	3	19	43	64	76	106	135	143	179	189
0.9	1	2	7	22	44	63	74	95	127	143	167
0.95	0	0	0	6	21	40	67	77	95	119	136
1	0	0	0	0	5	27	41	58	74	88	106

Threshold

Figure 4: Confusion matrix using *utility* alignment, *path* metric, and *translated* preprocess.

4.3 Effect of the Parameters

When analyzing the effect of **preprocess**, manual *translations* have a higher F-1 score than *raw* preprocessing in 1260/1428 cases, and a higher average F-1 score of 0.1386 compared to 0.1319 for *raw*. Our algorithm flagged just 39 out of 450 words (8.67%) for manual translations related to spelling errors, jargon, and acronyms. After performing these simple translations, we improved the F-1 score in 88.2% of cases and increased it on average by 5.12%. For the **metric** algorithm used through the WordNet API, the Wu-Palmer algorithm has higher F-1 score than the shortest path algorithm in 1143/1428 cases, as well as a higher average F-1 (0.157 vs. 0.113). Our results show that the Wu-Palmer metric is more suitable for aggregating causal maps in 80.0% of our configurations and increases the average F-1 score by 38.66%. Regarding the **alignment**, our utility matrix has a higher F-1 score than the standard approach in 1281/1428 cases, with an average F-1 score of 0.171 compared to 0.099. This parameter is responsible for even larger variation in F-1 score than *preprocess* or *alignment*, increasing it in 89.7% of cases and increasing the average F-1 score by 72.80%. These results **show that the Utility Matrix outperforms a traditional similarity matrix to align words** across causal maps when creating an aggregate conceptual model.

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Table 2: Configuration for interpreting individual parameters.

Parameter Configuration					
Preprocess	**Metric**	**Alignment**	α	**Threshold**	**Average F-1**
raw	*	*	*	*	0.1319
translated	*	*	*	*	0.1386
*	Shortest path	*	*	*	0.113
*	Wu-Palmer	*	*	*	0.157
*	*	standard	*	*	0.099
*	*	utility	*	*	0.171

Finally, out of the 24 combinations yielding the highest F-1 scores, 22 used $preprocess = translated$, $metric =$Wu-Palmer, and $alignment = utility$, with $0.80 \leq \alpha \leq 1.00$ and $0.450 \leq threshold \leq 0.650$. Because optimal values for α are generally close to 1.0, this shows that structural similarity is valuable for aggregating causal maps, but less so than semantic similarity.

5 DISCUSSION

The contributions of subject matter experts and/or the participation of community members are common activities to develop models of complex social systems. While many procedures have been thoroughly tested to elicit and represent knowledge from individuals, less research has been conducted on *combining* individual perspectives to create a conceptual model. Several of the available approaches are applicable mostly to ontologies, which have features (e.g., properties attached to concept nodes) that may not be available in the causal maps often constructed in preparation for simulation models (e.g., system dynamics, fuzzy cognitive maps). In this work, we used the combination of structural and semantic similarities from ontology matching and applied it to causal maps. Our methodological innovation rests in the novel use of a utility matrix, while our case study (serving to evaluate the performance of our methods) involves $n = 22$ previously collected maps alongside extensive tracking on their aggregation into one model.

Our two main takeaways are as follows. First, results demonstrate the potential of using an algorithm to aggregate causal maps. Although they contain less information than ontologies, they can still be systematically aggregated. Second, our results also confirm the potential value of our proposed utility matrix method, as it outperforms traditional methods on this case study.

Given the need to address concept names that were not recognized in WordNet, we also examined the benefit of a few curated human interventions in pre-processing the data and noted that it improved accuracy. However, reducing the necessity for such interventions would be an asset for modelers. Future works may thus explore the use of alternative databases to WordNet. One possibility is the Google Research API which can handle jargon and slang, as well as process languages other than English. Another avenue for future work would be to improve the structural metric, for example by matching nodes based on their position with respect to central concepts (Lavin et al. 2018). Finally, we would welcome further case studies to assess how performances depend on the domain of applications, or whether approaches should be tailored to characteristics of the participants' maps.

ACKNOWLEDGMENTS

Alexander J. Freund thanks the Department of Computer Science & Software Engineering and the Graduate School at Miami University for research funding.

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